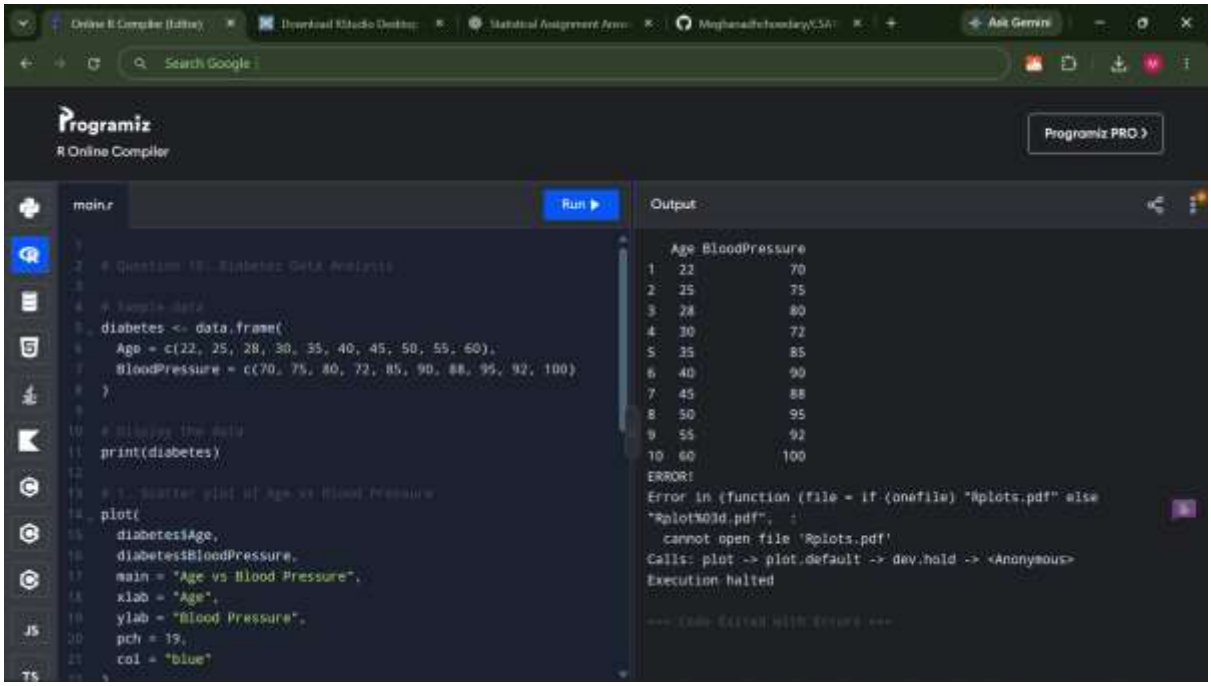
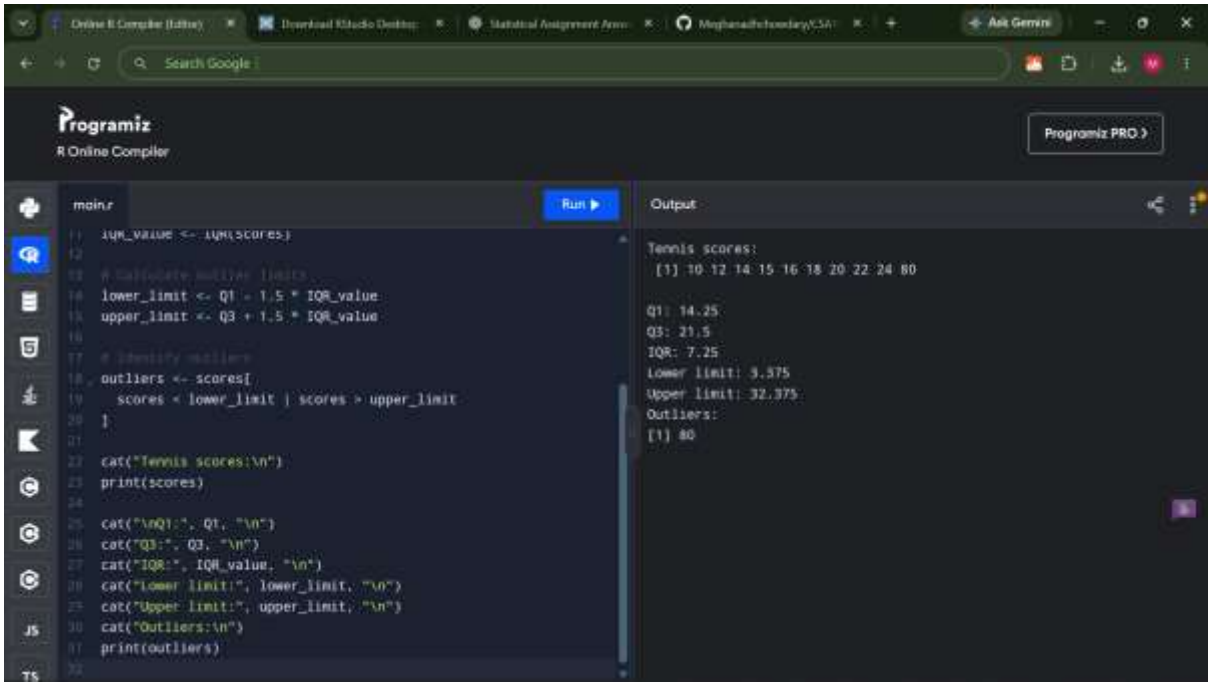
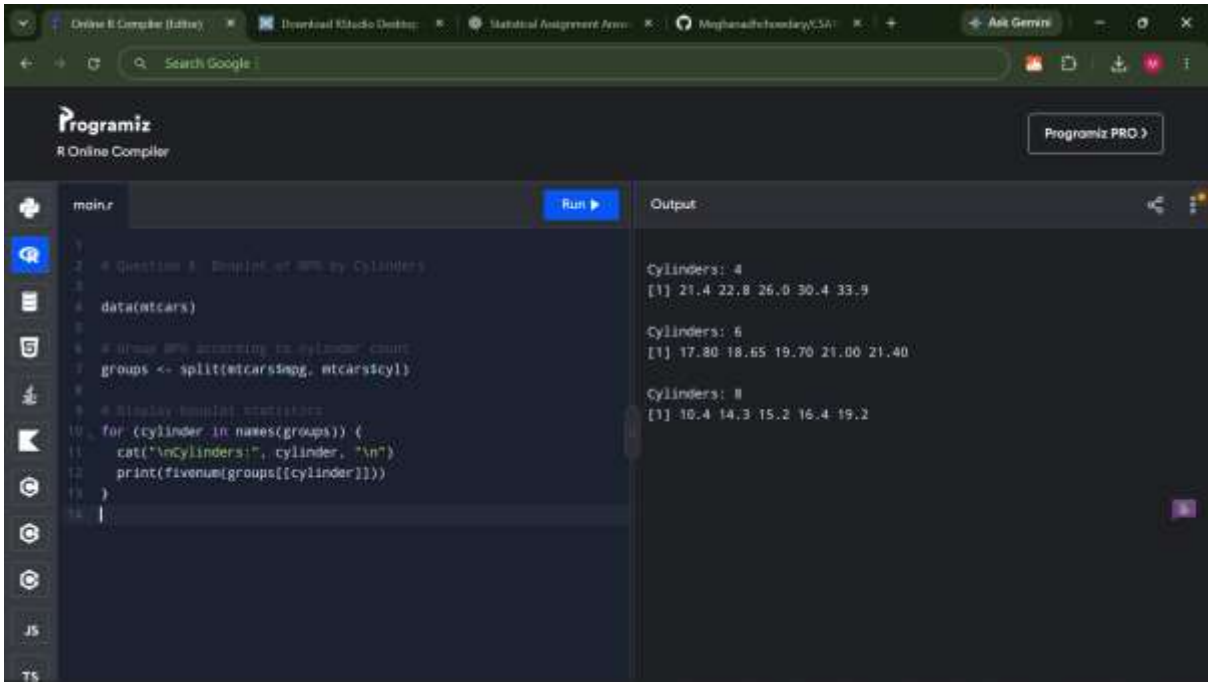
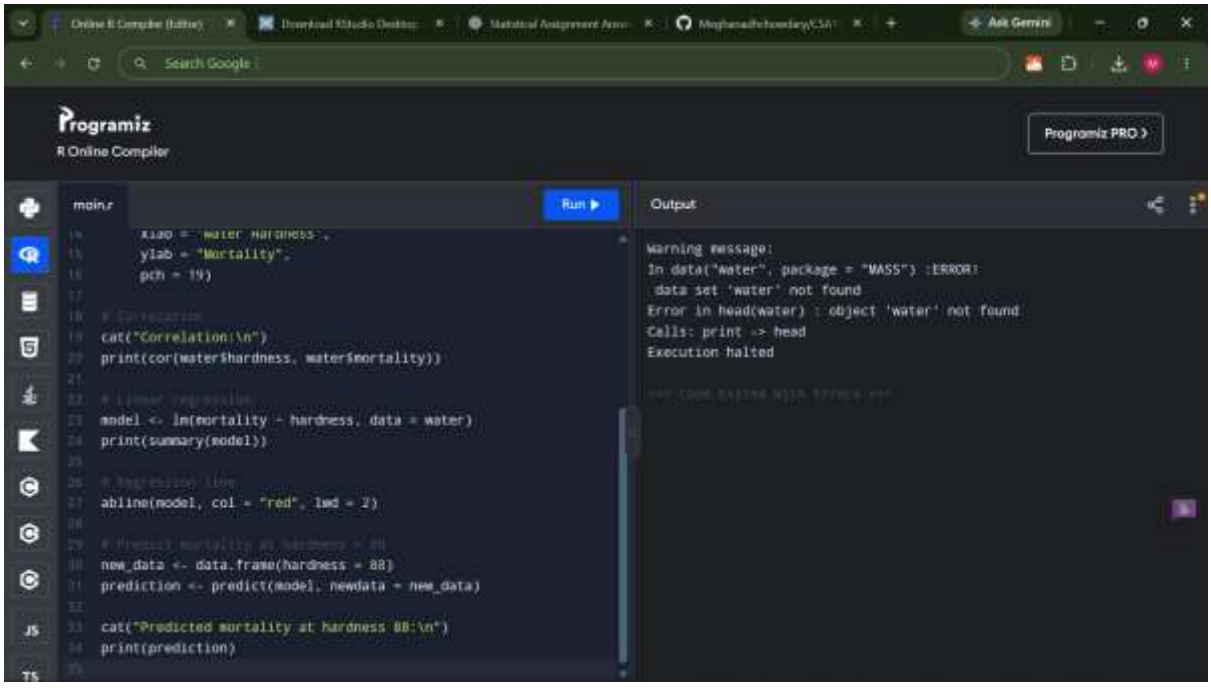
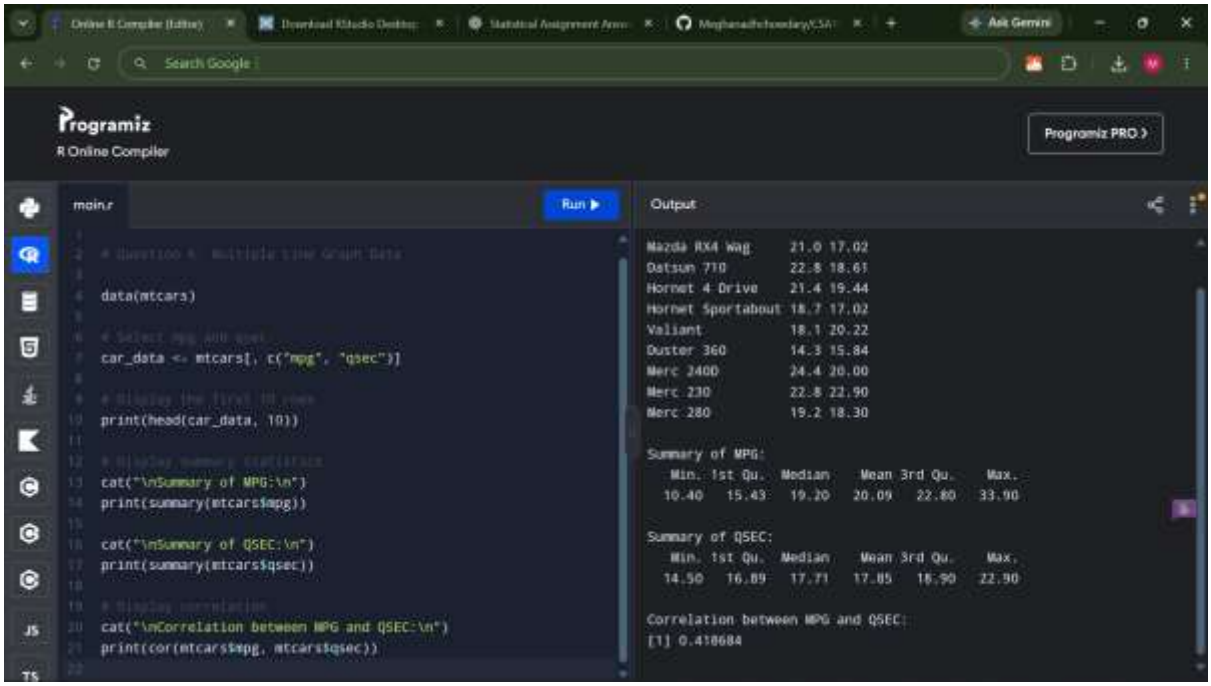
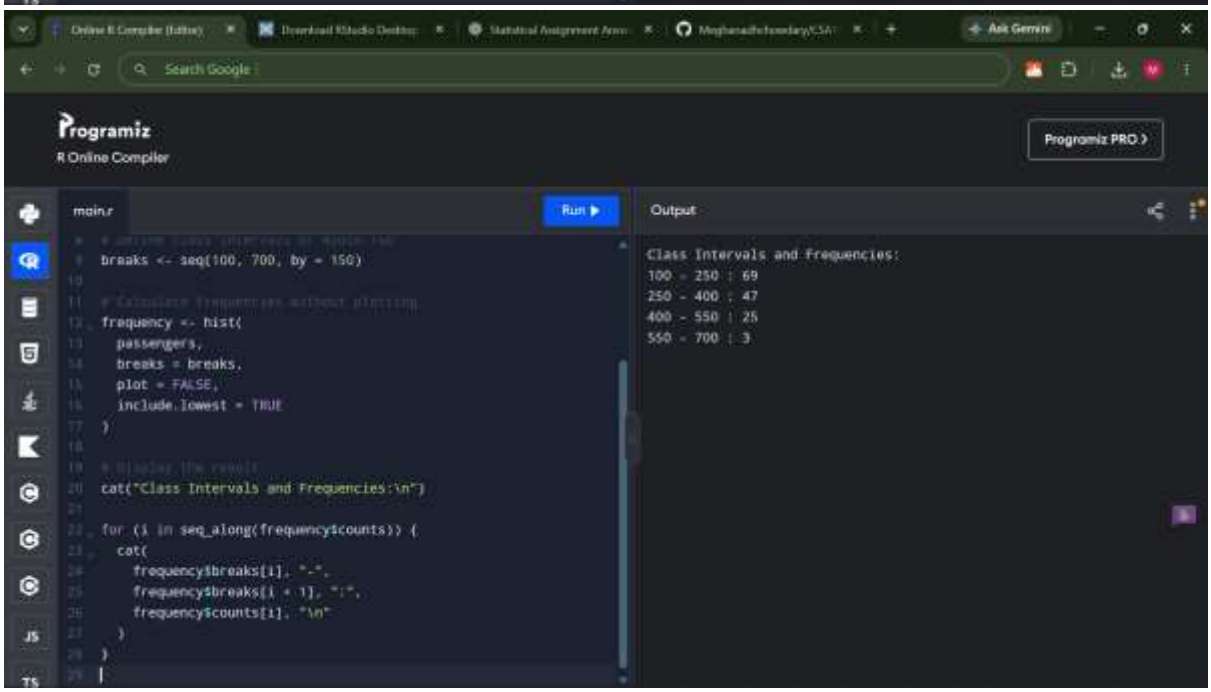

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Output
Question 10: Diabetes Data Analysis
Sample data
diabetes <- data.frame(
Age = c(22, 25, 28, 30, 35, 40, 45, 50, 55, 60),
BloodPressure = c(70, 75, 80, 72, 85, 90, 88, 95, 92, 100)
)
Display the data
print(diabetes)
1. Scatter plot of Age vs Blood Pressure
plot(
diabetes$Age,
diabetes$BloodPressure,
main = "Age vs Blood Pressure",
xlab = "Age",
ylab = "Blood Pressure",
pch = 19,
col = "blue"
Age BloodPressure
1 22 70
2 25 75
3 28 80
4 30 72
5 35 85
6 40 90
7 45 88
8 50 95
9 55 92
10 60 100
ERROR!
Error in (function (file = if (onefile) "Rplots.pdf" else
"Rplot%03d.pdf",
cannot open file 'Rplots.pdf'
Calls: plot -> plot.default -> dev.hold -> <Anonymous>
Execution halted
=== Code Exited with Errors ===

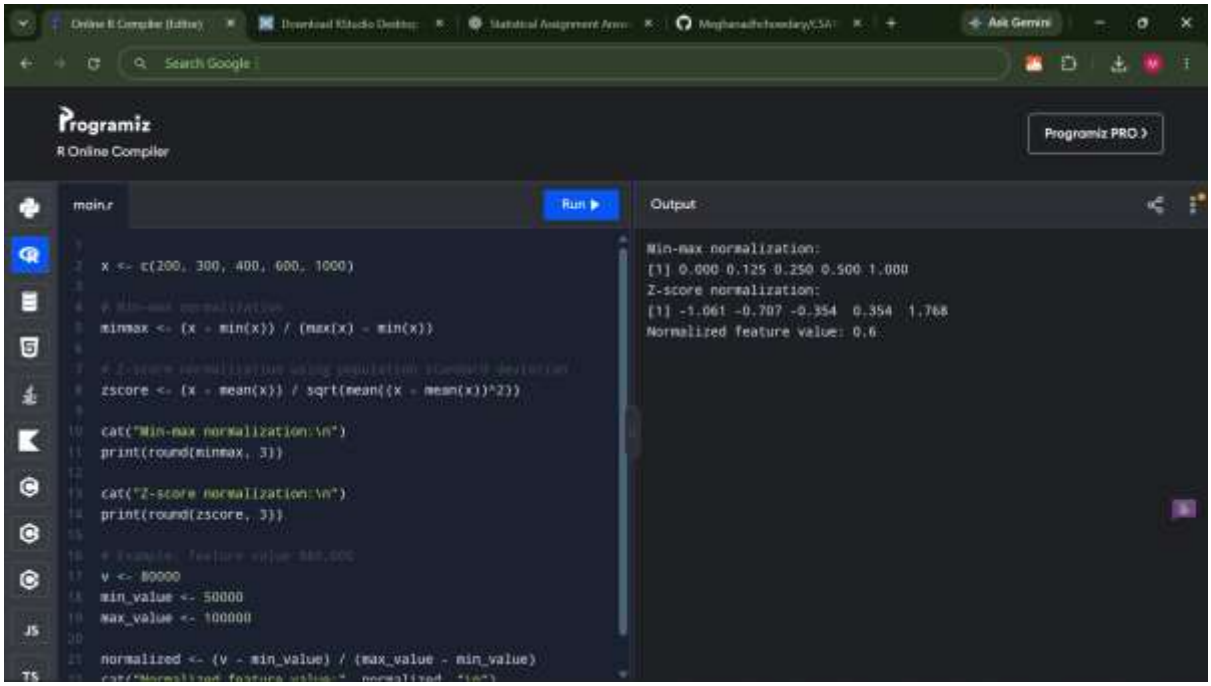

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IQR_value <- IQR(scores)
Calculate outlier limits
lower_limit <- Q1 - 1.5 * IQR_value
upper_limit <- Q3 + 1.5 * IQR_value
Identify outliers
outliers <- scores[
scores < lower_limit | scores > upper_limit
]
cat("Tennis scores:\n")
print(scores)
cat("\nQ1:", Q1, "\n")
cat("Q3:", Q3, "\n")
cat("IQR:", IQR_value, "\n")
cat("Lower limit:", lower_limit, "\n")
cat("Upper limit:", upper_limit, "\n")
cat("Outliers:\n")
print(outliers)
Tennis scores:
[1] 10 12 14 15 16 18 20 22 24 80
Q1: 14.25
Q3: 21.5
IQR: 7.25
Lower limit: 3.375
Upper limit: 32.375
Outliers:
[1] 80

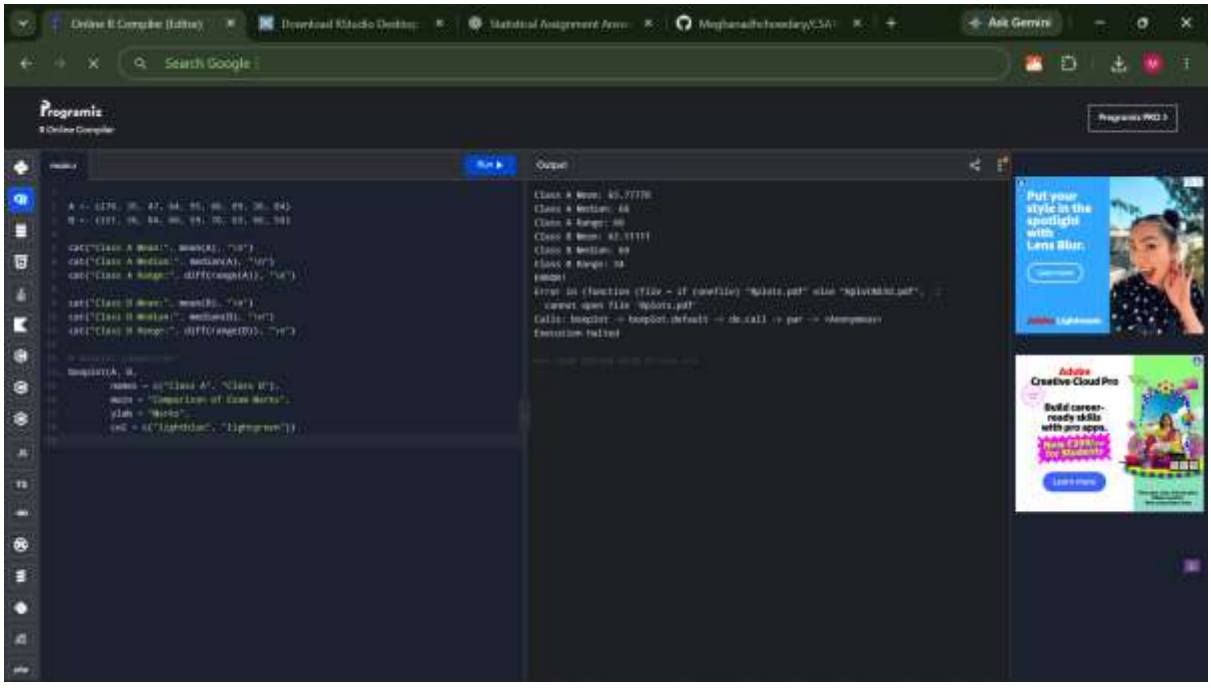

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Run
Output
Question 8: Boxplot of MPG by Cylinders
data(mtcars)
Group MPG according to cylinder count
groups <- split(mtcars$mpg, mtcars$cyl)
Display boxplot statistics
for (cylinder in names(groups)) {
cat("\nCylinders:", cylinder, "\n")
print(fivenum(groups[[cylinder]]))
}
Cylinders: 4
[1] 21.4 22.8 26.0 30.4 33.9
Cylinders: 6
[1] 17.80 18.65 19.70 21.00 21.40
Cylinders: 8
[1] 10.4 14.3 15.2 16.4 19.2

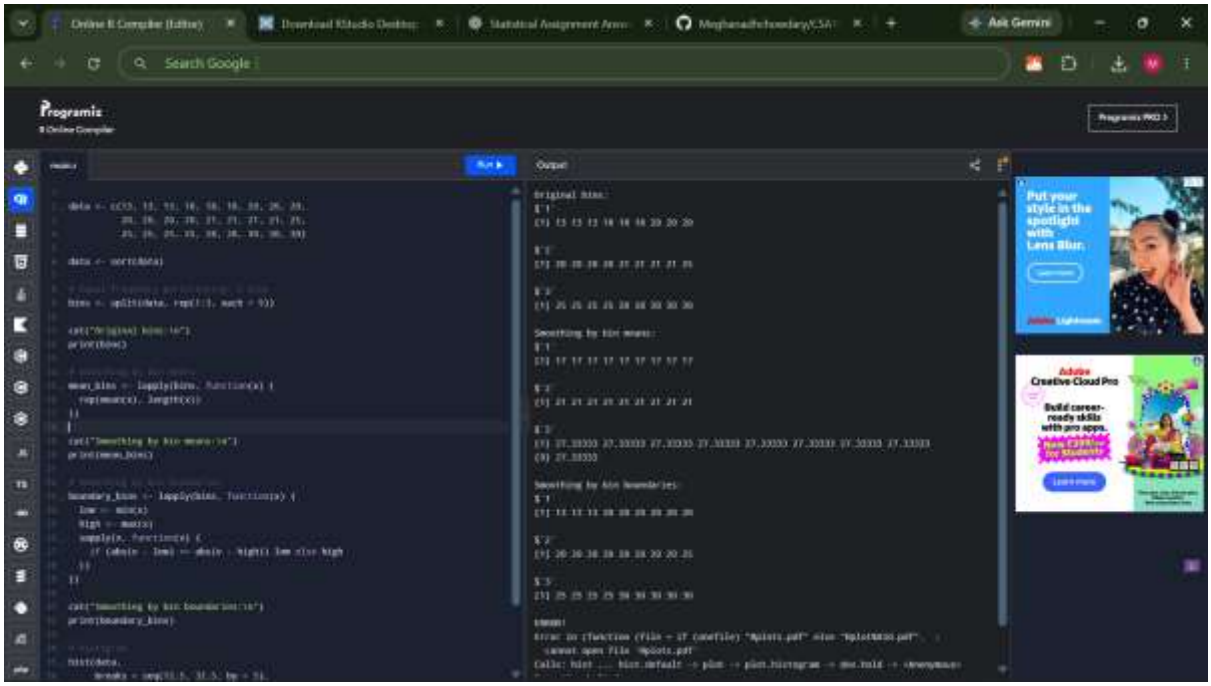

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Run
Output
xlab = "Water Hardness",
ylab = "Mortality",
pch = 19)
Correlation
cat("Correlation:\n")
print(cor(water$hardness, water$mortality))
Linear regression
model <- lm(mortality ~ hardness, data = water)
print(summary(model))
Regression line
abline(model, col = "red", lwd = 2)
Predict mortality at hardness = 88
new_data <- data.frame(hardness = 88)
prediction <- predict(model, newdata = new_data)
cat("Predicted mortality at hardness 88:\n")
print(prediction)
Warning message:
In data("water", package = "MASS") :ERROR:
data set 'water' not found
Error in head(water) : object 'water' not found
Calls: print -> head
Execution halted

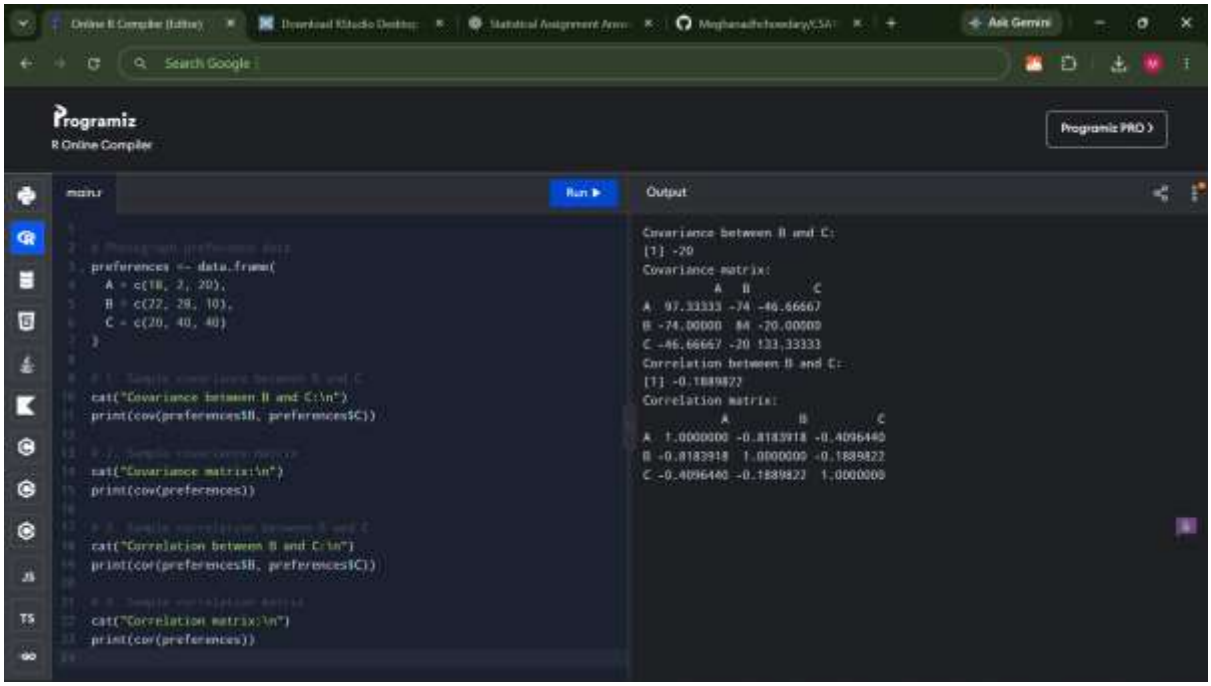

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Run
Output
Question 4 : Multiple Line Graph Data
data(mtcars)
Select mpg and qsec
car_data <- mtcars[, c("mpg", "qsec")]
Display the first 10 rows
print(head(car_data, 10))
Display summary statistics
cat("\nSummary of MPG:\n")
print(summary(mtcars$mpg))
cat("\nSummary of QSEC:\n")
print(summary(mtcars$qsec))
Display correlation
cat("\nCorrelation between MPG and QSEC:\n")
print(cor(mtcars$mpg, mtcars$qsec))
Mazda RX4 Wag 21.0 17.02
Datsun 710 22.8 18.61
Hornet 4 Drive 21.4 19.44
Hornet Sportabout 18.7 17.02
Valiant 18.1 20.22
Duster 360 14.3 15.84
Merc 240D 24.4 20.00
Merc 230 22.8 22.90
Merc 280 19.2 18.30
Summary of MPG:
Min. 1st Qu. Median Mean 3rd Qu. Max.
10.40 15.43 19.20 20.09 22.80 33.90
Summary of QSEC:
Min. 1st Qu. Median Mean 3rd Qu. Max.
14.50 16.89 17.71 17.85 18.90 22.90
Correlation between MPG and QSEC:
[1] 0.418684


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Output
breaks <- seq(100, 700, by = 150)
frequency <- hist(
passengers,
breaks = breaks,
plot = FALSE,
include.lowest = TRUE
)
cat("Class Intervals and Frequencies:\n")
for (i in seq_along(frequency$counts)) {
cat(
frequency$breaks[i], "-",
frequency$breaks[i + 1], ":",
frequency$counts[i], "\n"
)
}
Class Intervals and Frequencies:
100 - 250 : 69
250 - 400 : 47
400 - 550 : 25
550 - 700 : 3


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Output
x <- c(200, 300, 400, 600, 1000)
minmax <- (x - min(x)) / (max(x) - min(x))
zscore <- (x - mean(x)) / sqrt(mean((x - mean(x))^2))
cat("Min-max normalization:\n")
print(round(minmax, 3))
cat("Z-score normalization:\n")
print(round(zscore, 3))
v <- 80000
min_value <- 50000
max_value <- 100000
normalized <- (v - min_value) / (max_value - min_value)
Min-max normalization:
[1] 0.000 0.125 0.250 0.500 1.000
Z-score normalization:
[1] -1.061 -0.707 -0.354 0.354 1.768
Normalized feature value: 0.6


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Covariance between B and C:
[1] -20
Covariance matrix:
A B C
A 97.33333 -74 -46.66667
B -74.00000 84 -20.00000
C -46.66667 -20 133.33333
Correlation between B and C:
[1] -0.1889822
Correlation matrix:
A B C
A 1.0000000 -0.8183918 -0.4096440
B -0.8183918 1.0000000 -0.1889822
C -0.4096440 -0.1889822 1.0000000